

DIABETIC RETINOPATHY DETECTION FROM EYE FUNDUS IMAGES

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Abstract— Diabetic Retinopathy (DR) merupakan komplikasi diabetes mellitus yang dapat menyebabkan kebutaan apabila tidak terdeteksi secara dini. Pemeriksaan retina secara manual memerlukan waktu dan tenaga ahli, sehingga diperlukan metode otomatis yang efisien. Penelitian ini mengusulkan pendekatan berbasis *Deep Learning* menggunakan *Convolutional Neural Network* (CNN) untuk mengklasifikasikan tingkat keparahan Diabetic Retinopathy pada citra fundus retina. Dataset yang digunakan berasal dari Kaggle (EyePACS) dengan lima kelas keparahan, yaitu No DR, Mild, Moderate, Severe, dan Proliferative DR. Metodologi penelitian meliputi tahap *preprocessing* citra, pelatihan model CNN, dan evaluasi performa model. Hasil eksperimen menunjukkan bahwa CNN sederhana mampu melakukan klasifikasi tingkat keparahan DR dengan performa yang baik, sehingga berpotensi digunakan sebagai sistem pendukung deteksi dini.

Keywords—*Diabetic Retinopathy, Deep Learning, Convolutional Neural Network, Fundus Image, Medical Image Classification*

I. INTRODUCTION (HEADING 1)

Diabetic Retinopathy (DR) is a serious complication of diabetes mellitus that can lead to blindness in people of productive age. According to the World Health Organization (WHO), more than 537 million people worldwide were living with diabetes in 2021, and this number is predicted to continue to increase significantly in the coming decade (WHO, 2021). In Indonesia, the prevalence of diabetes reached 10.9% according to the 2018 Basic Health Research (Riskesmas) report, and approximately one-third of diabetic patients are at risk of developing DR, which can lead to permanent vision impairment.

Early detection using fundus images is crucial for preventing blindness, but manual examination is time-consuming, expensive, and requires limited expertise. Therefore, deep learning-based approaches, particularly convolutional neural networks (CNNs), have been widely studied to improve the accuracy and efficiency of automated DR detection.

Previous research has shown that CNNs are capable of extracting important features from fundus images and achieving high performance on DR classification tasks. A systematic review by Kassani & Kassani (2019)

demonstrated that CNN models can significantly improve the accuracy of DR detection compared to traditional methods. Another study by Muhammad et al. (2022) demonstrated that deep learning models can automatically identify DR severity with competitive performance on public datasets. Furthermore, a recent study by Akram et al. (2023) demonstrated that transfer learning models such as ResNet and DenseNet are capable of superior results in multi-class DR classification.

This research focuses on developing a CNN that is easily understood by beginning students yet still performs well on the Diabetic Retinopathy fundus dataset from Kaggle. This approach aims to demonstrate the end-to-end process from image processing to disease classification using a deep learning model.

II. METHODOLOGY

This research methodology consists of several main stages: data collection, image processing, training a Convolutional Neural Network (CNN) model, model performance evaluation, and system implementation. This method flow is designed to build an automatic Diabetic Retinopathy (DR) severity detection model from retinal images.

A. Data Collection

The dataset used in this study comes from the public resource Kaggle (Blindness Detection Dataset). The Kaggle EyePACS dataset contains a total of approximately 88,702 retinal fundus images. The retinal images in the dataset come from various cameras and settings with varying resolutions. The dataset will be categorized into five classes: 0 (No DR), 1 (Mild), 2 (Moderate), 3 (Severe), and 4 (Proliferative DR). The dataset can be accessed at:

<https://www.kaggle.com/competitions/diabetic-retinopathy-detection>

In the EyePACS dataset, retinal fundus images are provided separately for the left and right eyes of each patient. In this study, each eye image is treated as an independent data sample, following the original dataset annotation protocol.

The Diabetic Retinopathy label is assigned per eye image, not per patient. Therefore, it is possible for one eye to be diagnosed with DR while the other eye remains normal. In

such cases, each eye retains its original label without aggregation.

For example, if a patient's left eye is labeled as Moderate DR and the right eye is labeled as No DR, both images are included in the dataset as separate samples with their respective labels. This approach aligns with common practices in retinal image analysis and avoids introducing bias from patient-level label merging.

B. Image Preprocessing

This preprocessing stage is necessary because retinal images have significant variations in lighting, size, and noise. The preprocessing steps used in this study include:

1. Resizing the image to 224 x 224 pixels

To ensure the input is consistent with the CNN architecture.

2. Pixel normalization (0-1 or -1-1)

To improve training stability and accelerate model convergence.

3. Contrast Limited Adaptive Histogram Equalization (CLAHE)

Used to enhance the contrast of retinal structures such as blood vessels and microaneurysms.

4. Data Augmentation

Performed to reduce overfitting. The techniques used are:

- Rotation ($\pm 20^\circ$)
- Horizontal & vertical flip
- Random brightness
- Zoom (0.8-1.2)

Augmentation is important given the highly imbalanced class distribution in the dataset.

This preprocessing stage significantly helps the model recognize pathological features such as:

• Microaneurysm

Explanation: Small swellings in the walls of retinal blood vessels as they weaken due to high blood sugar.

Shape: Small red dots, usually the earliest sign of DR.

• Hemorrhage

Explanation: Bleeding in the retina due to ruptured blood vessels.

Shape: Dot/blot hemorrhage (round, larger than a microaneurysm), flame hemorrhage (elongated, flame-like).

• Neovascularization

Explanation: Abnormal, weak growth of new blood vessels. This occurs when the retina is deprived of oxygen.

Shape: A network of thin, small, branching blood vessels resembling plant roots.

C. CNN Model Training and Deep Learning Algorithm

This study employs a Deep Learning approach using Convolutional Neural Networks (CNN) to automatically classify the severity level of Diabetic Retinopathy from retinal fundus images. CNN is a widely used deep learning algorithm for image-based tasks due to its ability to automatically extract spatial and hierarchical features from images.

The CNN model used has a simple and efficient architecture, consisting of:

1. Input Layer: RGB retinal images resized to $224 \times 224 \times 3$
2. Three Convolutional Blocks:
 - Convolutional layers with increasing filters (32, 64, 128)
 - ReLU activation function
 - Max-pooling layers for spatial dimensionality reduction
3. Fully Connected Layer:
 - Dense layer with ReLU activation
 - Dropout layer to reduce overfitting
4. Output Layer:
 - Softmax activation with five output classes representing DR severity level.

D. Model Evaluation

The evaluation was conducted using various metrics commonly used in multi-class classification, namely:

1. Accuracy

2. Precision, Recall, F1-Score per class

3. Confusion Matrix: shows the distribution of correct and incorrect predictions for each DR severity level.

4. AUC (Area Under Curve): used to evaluate the model's ability to distinguish between healthy and DR classes.

In addition, training loss and validation loss were monitored to identify overfitting.

The dataset was divided into three subsets to support model training and performance evaluation.

In this study, the commonly used split ratio is applied:

- 70% for training
- 15% for validation
- 15% for testing

The training set is used to update the CNN weights, the validation set monitors model generalization during training, and the test set provides the final unbiased performance evaluation.

Flowchart:

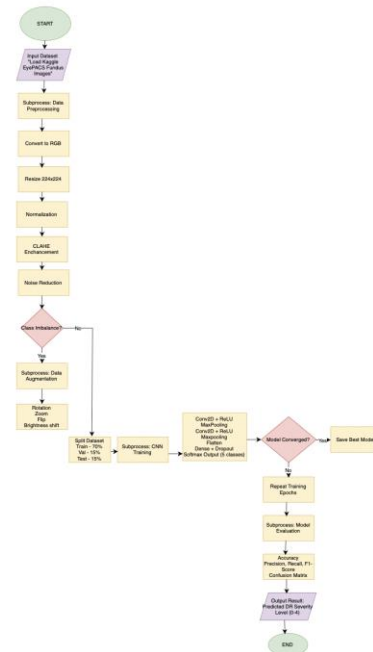


Fig. 1. Flowchart of the proposed cataract early detection method.

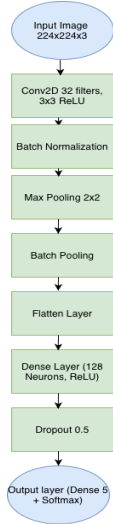


Fig. 2. Flowchart of the proposed CNN architecture.

III. EXPERIMENT AND RESULTS

A. Experimental Set-up

This experiment evaluates the performance of several Convolutional Neural Network (CNN) and ResNet-based architectures for diabetic retinopathy (DR) classification. All studies and experiments were conducted on the Kaggle EyePACS dataset, which contains 35,126 fundus images. Model training was performed using a GPU-based environment due to the high computational demands of deep learning architectures. The dataset was divided into training and testing sets to prevent information leakage and ensure objective model evaluation.

B. Dataset Characteristics

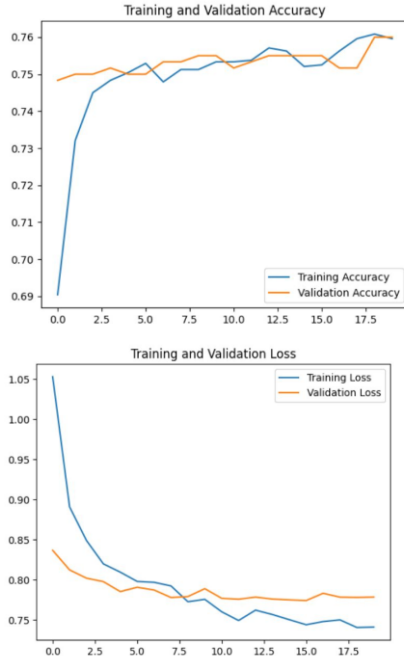


Fig. 3. Train-Validation Split Distribution

Percentage of Disease Severity (Pie Chart)

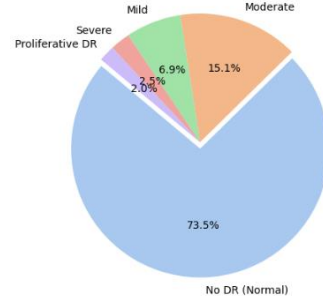


Fig. 4. Pie Chart of DR Severity

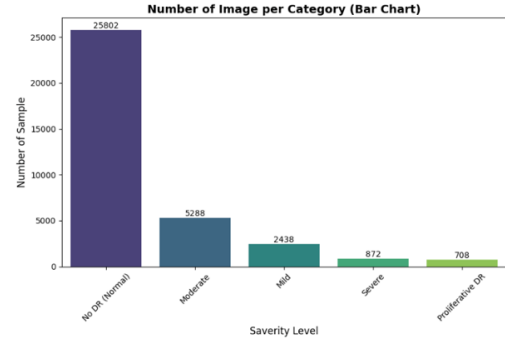


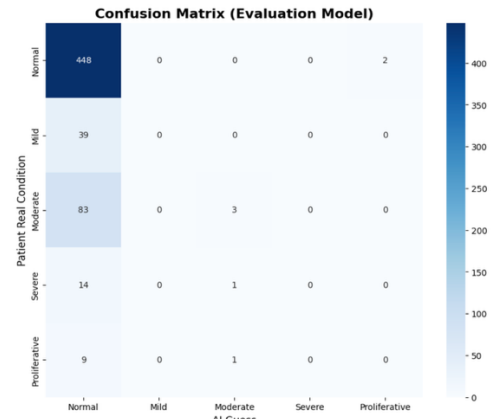
Fig. 5. Bar Chart of Image Count per Category

C. Preprocessing

Preprocessing steps include image resizing (typically 224×224), pixel normalization, and data augmentation to increase training diversity. Some studies also regrouped classes into *referable* versus *non-referable* DR to mitigate imbalance and improve model sensitivity toward diseased cases.

D. Evaluation Metrics

Model performance was assessed using accuracy, precision, recall, F1-score, and confusion matrix analysis. Figure 6 shows the confusion matrix generated from the evaluation, illustrating class-wise performance and common misclassification patterns. Figure 7 presents the classification



report, displaying precision, recall, and F1-score for each class. These metrics provide a deeper understanding of how well the model performs on both majority and minority classes.

Fig. 6. Confusion Matrix

=== AI GRADE REPORT (Classification Report) ===				
	precision	recall	f1-score	support
Normal	0.76	1.00	0.86	450
Mild	0.00	0.00	0.00	39
Moderate	0.60	0.03	0.07	86
Severe	0.00	0.00	0.00	15
Proliferative	0.00	0.00	0.00	10
accuracy			0.75	600
macro avg	0.27	0.21	0.18	600
weighted avg	0.65	0.75	0.65	600

Fig. 7. Classification Report (precision/recall/F1)

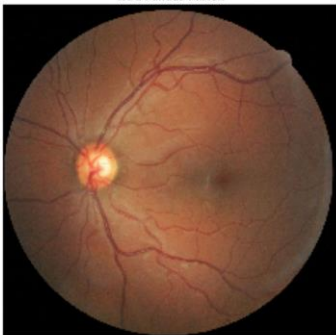
E. Patient Data Input Form

The proposed system provides a patient data input form that allows users to enter basic patient information and upload a retinal fundus image. The form includes fields for patient identification, such as the patient's name, which is used solely for data labeling and result organization.

Additionally, the user uploads an eye fundus image that serves as the primary input for diabetic retinopathy detection. Once submitted, the input data are forwarded to the preprocessing and classification stages of the system.

```
=====
FORM INPUT DATA PASIEN
=====
Nama Pasien      : Divika
ID/No. Rekam Medis: 12345
Usia             : 88
Dokter Pemeriksa : Tep

Citra Fundus Pasien
```



```
=====
LAPORAN DIAGNOSIS RETINOPATI
=====
Tanggal Periksa : 2026-01-04 12:07:19

Nama Pasien      : Divika
ID Pasien       : 12345
Usia            : 88
Dokter          : Tep

HASIL: Normal
Tingkat Keyakinan AI: 68.85%
```

F. Error Analysis

The model performs best in class 0 (No DR), corresponding to the majority class in the dataset. Most errors occur between Mild and Moderate DR due to subtle visual differences in retinal features and inconsistent image quality (blur, underexposure, glare). This is reflected in the confusion matrix (Fig. 4), where adjacent severity classes exhibit overlapping classification behavior.

G. Discussion

Overall, CNN and ResNet architectures show excellent effectiveness in detecting diabetic retinopathy, with many studies achieving accuracy above 90% and some reaching near-perfect results. However, these high values are influenced by the imbalanced nature of the EyePACS dataset.

Improving model generalization requires more balanced datasets, advanced augmentation, and evaluation across multiple data sources.

Unlike many previous studies that focus on complex deep learning architectures such as ResNet, DenseNet, or

ensemble models to achieve very high accuracy, this research emphasizes the development of a simple and lightweight CNN architecture that is suitable for educational and early screening purposes.

The main contributions of this study are:

1. Educational-oriented CNN design: The proposed CNN architecture is intentionally kept simple to demonstrate the complete workflow of diabetic retinopathy detection for beginner-level deep learning learners, while still maintaining reasonable performance.
2. Clinical feature-oriented preprocessing: The integration of Contrast Limited Adaptive Histogram Equalization (CLAHE) enhances clinically relevant retinal features such as microaneurysms, hemorrhages, and neovascularization, which are crucial indicators of DR severity.
3. Multi-class severity classification: Unlike studies that simplify the problem into binary classification (DR vs. No DR), this research performs five-class DR severity classification, which better reflects real clinical grading scenarios.

Therefore, this study bridges the gap between highly complex research models and practical, interpretable deep learning implementations suitable for academic learning and early-stage screening systems.

IV. CONCLUSION

This research successfully implemented a Convolutional Neural Network (CNN) deep learning model to detect and classify the severity of Diabetic Retinopathy (DR) using retinal fundus images from Kaggle EyePACS dataset. The study highlights that preprocessing stages, specifically resizing, normalization, Contrast Limited Adaptive Histogram Equalization (CLAHE), and data augmentation, are critical in handling significant lighting variations and class imbalances inherent in medical datasets.

The experimental results demonstrate that the proposed simple CNN architecture is capable of classifying DR severity with an overall accuracy of 75%. While this indicates good potential for a lightweight model, the performance varies significantly across classes due to the highly imbalanced dataset. The model performs well on the majority class (No DR) but faces challenges in correctly identifying minority classes such as Mild and Severe DR, as reflected in the lower recall scores for these categories. This confirms that distinguishing between adjacent severity levels remains a complex task due to subtle visual differences in retinal features.

In conclusion, while the current model shows promise as an initial screening tool, further improvements are necessary for clinical reliability. Future work should focus on addressing class imbalance through advanced techniques like oversampling or synthetic data generation (GANs) and testing on more diverse datasets. These steps are crucial to enhance the model's sensitivity to early-stage disease and ensure its effectiveness as a robust decision support system for preventing blindness.

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